

The image shows a sequence of four Feynman diagrams representing a four-point function with two vertical red lines. The diagrams are summed together and then equated to an integral expression.

The first diagram shows a four-point function with external momenta  $p_1$  and  $p_2$  on both sides of a vertical red line. The internal lines are a fermion loop and a gluon line. The second diagram shows a similar four-point function with a different internal structure. The third diagram shows a four-point function with a different internal structure. The fourth diagram shows a four-point function with a different internal structure.

The sum of these four diagrams is equal to the following integral expression:

$$= \text{disc}_{p_1^2} \text{disc}_{p_2^2} \int \frac{d^d k}{(2\pi)^d} \frac{N_{\boxtimes}^{q\bar{q}} \delta_{q^2}^+(p_{12} - k) \delta_0^+(k)}{p_1^2 p_2^2 (k - p_2)^2 (k - p_1)^2}$$